

Standard-dose unfractionated heparin versus low-dose unfractionated heparin and low-molecular-weight heparin in extracorporeal life support (RATE): an open-label, randomised, non-inferiority trial



Olivier van Minnen, Annemieke Oude Lansink-Hartgring, Rombout B E van Amstel, Bas van den Bogaard, Jeroen J H Bunge, Thijs S R Delnoij, Joep M Droogh, Laurien van Koppenhagen, Carlos V Elzo Kraemer, Marijn Kuijpers, Jacinta J Maas, Jesse de Metz, Marcel C G van de Poll, S Jorinde Raasveld, Dinis dos Reis Miranda, Priya Vart, Karin M Vermeulen, Alexander P J Vlaar, Walter M van den Bergh, on behalf of the Dutch ECLS Study Group*

Summary

Background In patients receiving extracorporeal membrane oxygenation (ECMO), standard practice is full-dose intravenous unfractionated heparin (UFH) targeting an activated partial thromboplastin time of 2·0–2·5 times baseline to reduce thrombotic risk. This approach can increase bleeding without further reducing thrombosis compared with low-dose UFH. Robust evidence to guide anticoagulation targets is absent because anticoagulation targets in ECMO have never been assessed in a sufficiently powered randomised trial. We aimed to determine whether low-dose UFH or therapeutic low-molecular-weight heparin (LMWH) is non-inferior to standard-dose UFH in patients receiving ECMO.

Methods In this open-label, three-arm, randomised, non-inferiority trial at seven Dutch intensive care units (ICUs), adults aged 18 years or older supported with veno-venous or veno-arterial ECMO at the ICU without vital indication for full-dose anticoagulation (eg, mechanical mitral valve) were randomly assigned to intravenous standard-dose UFH (activated partial thromboplastin time 2·0–2·5 times baseline), intravenous low-dose UFH (1·5–2·0 times baseline), or therapeutic subcutaneous LMWH. The primary outcome was a composite of severe bleeding during ECMO support, severe thromboembolic complications during ECMO support, or all-cause mortality at 6 months, and was assessed in all randomly assigned patients with deferred informed consent and 6-month follow-up data according to the intention-to-treat principle. Non-inferiority was met if the upper 95% CI limit for the absolute risk difference was less than 7·5 percentage points. This trial is registered at ClinicalTrials.gov (NCT04536272) and the Dutch trial register (NL7976).

Findings Between Oct 22, 2020, and Sept 12, 2024, 330 patients were enrolled: 110 were randomly assigned to standard-dose UFH, 110 to low-dose UFH, and 110 to LMWH. Of 330 enrolled patients, 320 (225 [70%] males, 95 [30%] females; median age 56 years [IQR 45–65]; 255 [80%] White ethnicity) were analysed at 6 months. The composite primary outcome occurred in 87 (81%) of 107 patients with standard-dose UFH, 78 (72%) of 108 with low-dose UFH (absolute risk difference –9·1 percentage points [95% CI –20·3 to 2·1]), and 79 (75%) of 105 with LMWH (–6·1 percentage points [–17·2 to 5·0]), meeting non-inferiority for both interventions. The frequency of severe bleeding was lower with low-dose UFH and LMWH than with standard-dose UFH (63 [58%] patients and 62 [59%] vs 70 [65%]), without excess severe thromboembolic complications (11 [10%] and nine [9%] vs 12 [11%]), although these differences did not reach statistical significance. At 6 months, 54 (50%) patients in the standard-dose UFH group, 45 (42%) in the low-dose UFH group, and 46 (44%) in the LMWH group had died.

Interpretation Low-dose UFH and therapeutic LMWH were non-inferior to standard-dose UFH. This finding supports reconsideration of anticoagulation targets in ECMO. Given the global expansion of ECMO, these findings suggest that reduced anticoagulation targets could substantially reduce bleeding-related harm.

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Introduction

Extracorporeal membrane oxygenation (ECMO) provides temporary circulatory or respiratory support in selected patients with severe cardiac or pulmonary failure, refractory to alternative therapies. Despite technological advances and expanding use, mortality

remains high, ranging from 34% to 50% with veno-venous ECMO and 50% to 70% with veno-arterial ECMO.^{1–5} A substantial proportion of morbidity, however, arises from treatment-related complications, particularly bleeding and thrombosis. Bleeding complications are far more common and more strongly

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*Members are listed at the end of the Article

Department of Critical Care (O van Minnen MD, A Oude Lansink-Hartgring MD PhD, J M Droogh MD PhD, W M van den Bergh MD PhD), **Department of Anesthesiology** (O van Minnen), and **Department of Clinical Pharmacy and Pharmacology** (P Vart PhD), University Medical Center Groningen, Groningen, Netherlands; **Department of Intensive Care Medicine** (R B E van Amstel MD PhD, S J Raasveld MD PhD, Prof A P J Vlaar MD PhD) and **Department of Anesthesiology** (S J Raasveld), Amsterdam University Medical Center, Amsterdam, Netherlands; **Department of Intensive Care, OLVG, Amsterdam, Netherlands** (B van den Bogaard MD PhD, J de Metz MD PhD); **Department of Intensive Care Medicine** (J J H Bunge MD PhD, D dos Reis Miranda MD PhD) and **Department of Cardiology** (J J H Bunge), Erasmus Medical Center, Rotterdam, Netherlands; **Department of Intensive Care Medicine** (T S R Delnoij MD PhD, M C G van de Poll MD PhD) and **Department of Surgery** (M C G van de Poll), Maastricht University Medical Center, Maastricht, Netherlands; **Department of Intensive Care Medicine, Isala Clinics, Zwolle, Netherlands** (L van Koppenhagen MD PhD, M Kuijpers MD PhD);

Department of Intensive Care
Medicine, Leiden University
Medical Center, Leiden,
Netherlands

(C V Elzo Kraemer MD,
J J Maas MD PhD); School for
Nutrition and Translational
Research in Metabolism,
Maastricht University,
Maastricht, Netherlands
(M C G van de Poll); Department
of Epidemiology, University of
Groningen, University Medical
Center Groningen, Groningen,
Netherlands

(K M Vermeulen PhD)

Correspondence to:
Olivier van Minnen, Department
of Critical Care, University
Medical Center Groningen,
9700 RB Groningen, Netherlands
o.van.minnen@umcg.nl

Research in context

Evidence before this study

We searched PubMed from inception to June 29, 2025, for studies on anticoagulation strategies in adult extracorporeal membrane oxygenation (ECMO) using terms including “ECMO”, “anticoagulation”, “heparin”, and “low-molecular-weight heparin”. Evidence consisted mainly of observational studies, registry analyses, and one small pilot randomised trial. These studies suggested that lower-intensity anticoagulation might reduce bleeding without increasing thrombosis, but were limited by small sample sizes, heterogeneity, and risk of bias. No adequately powered randomised controlled trials comparing anticoagulation intensity strategies in ECMO have been completed.

Added value of this study

This multicentre, randomised, non-inferiority trial provides the first adequately powered comparison of standard-dose

unfractionated heparin with low-dose unfractionated heparin and therapeutic low-molecular-weight heparin in ECMO. Both strategies were non-inferior to the standard dose for the composite endpoint consisting of severe bleeding, severe thromboembolic complications, and 6-month mortality, with consistent signals towards reduced bleeding and improved cost-effectiveness.

Implications of all the available evidence

The evidence supports reducing anticoagulation intensity in ECMO because lower-intensity strategies seem to be safe and might reduce bleeding complications and costs. These findings support reconsideration of current anticoagulation targets and might inform future guideline development and clinical decision making.

associated with mortality than clinically significant thrombosis during ECMO.^{6,7}

Exposure of blood to ECMO circuits activates coagulation, predisposing to thrombosis.⁸ To mitigate this issue, most centres administer full-dose unfractionated heparin (UFH) targeting an activated partial thromboplastin time (aPTT) of 2.0–2.5 times baseline (about 60–75 s), although this target has never been validated in adequately powered randomised trials.^{4,9–11}

Anticoagulation is intended to prevent thrombotic events. However, modern ECMO systems with centrifugal pumps and coated tubing have reduced patient-level thrombotic events to about 1–10%,^{10,12} whereas bleeding complications affect up to half of patients and more strongly affect mortality; studies reporting higher thrombotic rates predominantly reflect oxygenator-associated thrombosis rather than clinically manifest patient events.^{6,7,13} Higher anticoagulation targets are strongly associated with haemorrhagic complications and subsequent transfusion requirements, both of which are linked to worse outcomes.^{2,5,10,14–20}

Close monitoring of UFH is mandatory owing to its unpredictable pharmacokinetics and pharmacodynamics, but there is still no consensus on the optimal approach to UFH monitoring in clinical practice. Initial dosing and adjustment of UFH infusions rely on few data with low clinical relevance, which probably contributes to heterogeneity in practice. Despite all of these limitations, aPTT is the most widely used parameter to monitor UFH treatment.

Observational studies and a small feasibility study suggest that lower UFH targets might reduce haemorrhagic complications without increasing thrombotic complications.^{21–25} Data on low-molecular-weight heparin (LMWH), either prophylactic or therapeutic dosage, in ECMO remain scarce.^{21–25}

We aimed to determine whether low-dose UFH or therapeutic LMWH is non-inferior to standard-dose UFH for a composite of major bleeding, major thrombotic events, and 6-month mortality in adults receiving ECMO. We hypothesised that lower-intensity regimens would decrease bleeding and transfusion needs without increasing thrombosis or mortality, thereby improving quality of life and cost-effectiveness.

Methods

Study design and participants

We conducted an open-label, three-arm, randomised, non-inferiority trial at seven closed-format mixed intensive care units (ICUs) in the Netherlands. We enrolled adults aged 18 years or older receiving venovenous or veno-arterial ECMO in the ICU for severe respiratory or circulatory failure considered to be reversible by the treating team or initiated as a bridge to transplantation. Exclusion criteria were procedural ECMO use (eg, high-risk percutaneous coronary intervention or intraoperative support), absence of deferred consent, a compelling indication for full-dose anticoagulation (eg, mechanical mitral valve, proven pulmonary embolism, and intracardiac thrombus), or heparin-induced thrombocytopenia.

The study was approved by the Medical Ethics Review Board of the University Medical Center Groningen (Groningen, Netherlands; approval number: 2020.042). Deferred informed consent was obtained from patients or legal surrogates after the local investigator or research nurse informed the patients about the study intervention when they regained consciousness. The trial protocol is published.²⁶ The statistical analysis plan is available in the appendix (pp 20–36). This trial is registered at ClinicalTrials.gov (NCT04536272) and the Dutch trial register (NL7976).

See Online for appendix

Randomisation and masking

Patients were randomly assigned (1:1:1) to receive standard-dose UFH, low-dose UFH, or LMWH using a secure web-based system (ALEA, ALEA Clinical, Abcoude, Netherlands), with variable block sizes, stratified by study site and ECMO mode (veno-venous vs veno-arterial). Allocation was concealed until assignment. The trial was open label, but study personnel responsible for follow-up were masked to treatment allocation.

Procedures

According to random assignment, patients received standard-dose UFH administered as a continuous intravenous infusion, targeting an aPTT of 2·0–2·5 times the centre-specific baseline (about 60–75 s depending on local assay characteristics); low-dose UFH administered as a continuous intravenous infusion, targeting 1·5–2·0 times baseline (about 45–60 s); or subcutaneous therapeutic LMWH twice daily, adjusted for bodyweight and renal function. Anticoagulation was initiated after cannulation and could be deferred for up to 24 h if active bleeding was present (eg, postoperative or cannulation related). A UFH loading dose of 5000 IU was commonly given at percutaneous cannulation according to local practice. In the UFH groups, aPTT was measured four to six times daily, and UFH doses were adjusted according to local standards. aPTT assays were performed according to local laboratory protocols, and reagent variability between centres was not standardised. Anti-factor Xa activity was added to the case report forms midway through the trial, but not used for dose adjustment, and was measured once daily. Measurements were obtained once daily; for patients receiving UFH (continuous infusion), sampling was performed at 0600 h each morning, whereas for patients receiving LMWH (twice-daily dosing), each participating site collected anti-factor Xa samples about 4 h after dose administration per institutional routine to reflect peak activity. The anticoagulation study regimens were maintained during ECMO support only. Ethnicity data were collected from the medical record.

Outcomes

The primary outcome was a composite of severe bleeding, defined according to the Extracorporeal Life Support Organization Registry criteria (more than three units packed red blood cells or whole blood transfused in a calendar day, bleeding requiring intervention, or any CNS bleeding) during ECMO support; severe thromboembolic complications (ischaemic stroke, limb ischaemia, or acute pump failure) during ECMO support; and all-cause mortality at 6 months.²⁷ The composite primary outcome incorporated events occurring during ECMO support as well as 6-month mortality, reflecting the clinical rationale that anticoagulation strategies might affect both immediate device-related complications and downstream patient outcomes beyond decannulation, such as transfusion-related morbidity.

Secondary outcomes were the individual components of the primary outcome; transfusion of blood products; all membrane oxygenator exchanges; pulmonary embolism; all haemorrhagic or thromboembolic complications; days alive and free from mechanical ventilation, ECMO, and ICU at day 60; coagulation parameters (aPTT, prothrombin time, fibrinogen, haemoglobin, and platelets); health-related quality of life at 6 months; health-care costs; cost-effectiveness; reason for ECMO discontinuation; and discharge location. These outcomes were selected because they might be affected by the anticoagulation regimen and capture both patient-centred and societal impact.

Statistical analysis

We assumed a 70% event rate for the primary outcome in the standard-dose UFH group and 60% event rate in each intervention group. With a non-inferiority margin of 7·5 percentage points, 91 patients per group would provide about 80% power at a one-sided alpha of 2·5%. The non-inferiority margin was chosen a priori as the maximum clinically acceptable difference, ensuring that any potential loss in efficacy would be outweighed by anticipated reductions in bleeding and transfusion-related morbidity, given the low incidence of clinically disabling thrombotic events in ECMO. To ensure adequate power and allow for potential loss to follow-up, the target sample size was set at 330 patients.

For the primary outcome, all participants who gave delayed informed consent and had data for the outcome at 6 months were analysed according to their randomly assigned treatment group (ie, intention-to-treat principle), irrespective of subsequent changes in anticoagulation regimen or crossover between anticoagulation strategies. For the primary analysis, each intervention group was compared separately with the standard-dose UFH group. Absolute risk differences with two-sided 95% CIs were calculated. Relative risks (RRs) were estimated using Poisson regression with robust variance applied to binary outcomes, adjusted for ECMO mode and study site as prespecified stratification factors. Missing data were not imputed. We also performed a per-protocol analyses for the primary endpoint, including all patients without a crossover to another study group.

Secondary binary outcomes, which were analysed according to the intention-to-treat principle, were compared with χ^2 or Fisher's exact tests and expressed as RRs with 95% CIs. Continuous outcomes were analysed with ANCOVA or rank-based ANCOVA if distributions were skewed, adjusted for baseline values when appropriate. Days alive and free from mechanical ventilation, ECMO, and the ICU at day 60 were compared using the Kruskal–Wallis test. p values were calculated from comparisons between each intervention and the control group and are reported accordingly. Ordinal outcomes were analysed using rank-based methods.

Prespecified subgroup analyses were conducted for the primary composite outcome and its individual components

according to ECMO mode (veno-venous or veno-arterial), indication (pulmonary, cardiac, or extracorporeal cardiopulmonary resuscitation), age (above or below the median), surgical status (yes or no), cannulation method (central or peripheral), cannula (heparin-coated or non-heparin-coated), ECMO equipment (Cardiohelp [Getinge, Gothenburg, Sweden], Xenios [Heilbronn, Germany], or other), and COVID-19 status. No adjustment was made for multiple testing, and secondary and subgroup results should be considered as hypothesis generating.

Cost-effectiveness was assessed from a societal perspective over a 6-month horizon. Included costs were those related to ECMO treatment, hospital stay, blood products, medication, and productivity losses. Trial-based data were supplemented with resource use from our previous study.²⁸ Health outcomes were expressed as quality-adjusted life-years (QALYs) derived from EQ-5D-5L collected by telephone at 6 months and valued using the Dutch tariff.²⁹ For participants who died before the 6-month follow-up, we assumed a utility of zero, given the very poor health state in this population. For the QALY calculations, a baseline utility value of 0.88 was assumed based on our previous study. Costs were valued in 2024 euros using Dutch standard prices according to national costing guidelines.³⁰ ECMO-related costs were derived from published ECMO cost data and recalculated on a per-day basis.³¹ The actual ECMO duration for each patient was used to estimate individual ECMO-related costs, after which daily costs were indexed to 2024 values using the Consumer Price Index,³² resulting in an adjustment factor of 1.357. Medication-related costs included major cost drivers such as blood products (red

blood cells, platelets, fresh frozen plasma, and fibrinogen) and high-cost haemostatic therapies, including recombinant factor VIIa (eptacog alfa), tranexamic acid, and prothrombin complex concentrate (Cofact, Sanquin Plasma Products, Amsterdam, Netherlands). Incremental cost-effectiveness ratios (ICERs) were calculated, and uncertainty was assessed using 5000 bootstrap replications and presented in cost-effectiveness planes.

An independent safety committee reviewed unblinded safety data every 3 months and conducted the planned interim analysis after 150 patients had been enrolled; investigators remained masked to all interim results.

IBM SPSS Statistics for Windows (version 30.0.0.0) was used for statistical analyses.

Role of the funding source

The funder of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Results

Between Oct 22, 2020, and Sept 12, 2024, 330 patients were enrolled and randomly assigned: 110 patients were assigned to each group (figure 1; see appendix p 4 for inclusions per centre). A prespecified interim analysis was conducted after enrolment of 150 patients, after which the safety committee advised continuing the study. For the primary analysis at 6 months, data were available for 320 patients (225 [70%] males, 95 [30%] females; median age 56 years [IQR 45–65]; 255 [80%] White ethnicity): 107 in the standard-dose UFH group, 108 in the low-dose UFH group, and 105 in the LMWH group. Baseline characteristics are shown in table 1. The median age was 55 years (IQR 43–64) in the standard-dose UFH group, 56 years (45–64) in the low-dose UFH group, and 58 years (47–68) in the LMWH group. 46 (43%) patients in the standard-dose UFH group, 40 (37%) patients in the low-dose UFH group, and 42 (40%) patients in the LMWH group received veno-venous ECMO. The most common indications for ECMO were severe acute respiratory distress syndrome (93 [29%] of 320) and cardiogenic shock (192 [60%]). The mean ECMO duration was 6 days (SD 3) patients for supported with veno-arterial ECMO and 16 days (11) for patients supported with ven-venous ECMO.

Median daily lowest and highest aPTT values during ECMO support ranged from 48 s (IQR 43–54) to 68 s (60–77) in the standard-dose UFH group, 42 s (38–47) to 56 s (49–65) in the low-dose UFH group, and 34 s (30–40) to 38 s (33–50) in the LMWH group (appendix p 2). Median anti-factor Xa activity was 0.37 IU/mL (IQR 0.25–0.46) in the standard-dose UFH group, 0.25 IU/mL (0.19–0.49) in the low-dose UFH group, and 0.41 IU/mL (0.27–0.65) in the LMWH group (appendix p 2). Anti-factor Xa measurements were available for a subset of patients (n=71), as this parameter was incorporated into the case report forms midway through the trial. These results confirm a clear separation between the treatment groups (figure 2).

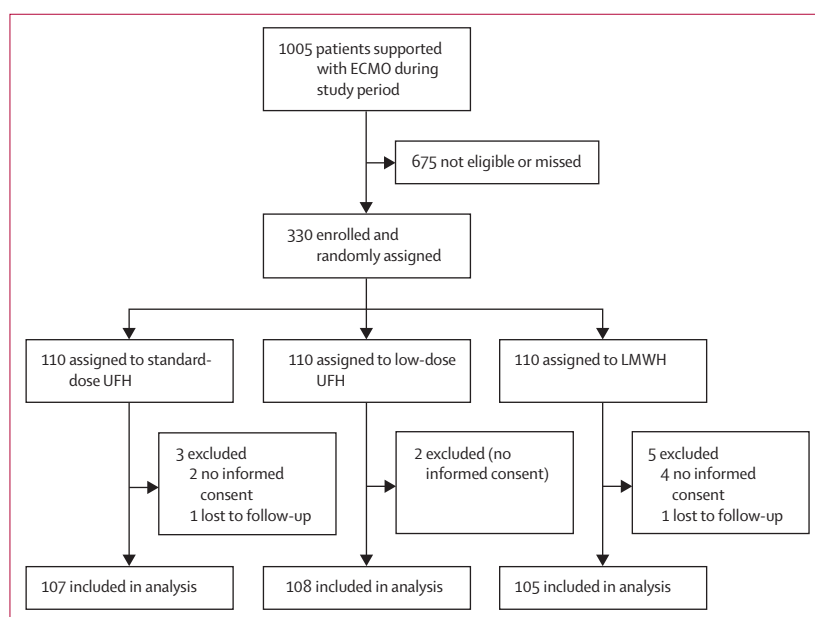


Figure 1: Trial profile

ECMO=extracorporeal membrane oxygenation. LMWH=low-molecular-weight heparin. UFH=unfractionated heparin.

There were no missing data for the primary outcome or for covariates included in the main model. At 6 months, the primary outcome occurred in 87 (81%) of 107 patients in the standard-dose UFH group, 78 (72%) of 108 patients in the low-dose UFH group, and 79 (75%) of 105 patients in the LMWH group. The absolute risk difference versus standard-dose UFH was -9.1 percentage points (95% CI -20.3 to 2.1) for low-dose UFH and -6.1 percentage points (-17.2 to 5.0) for LMWH. In both intervention groups, the upper bound of the 95% CI was below the non-inferiority margin of 7.5 percentage points, meeting the criterion for non-inferiority. Absolute risk differences with 95% CIs for the primary outcome and its components are shown in figure 3. After adjustment with the stratification factors centre and ECMO mode, results remained unchanged (appendix p 4).

Severe bleeding occurred in 70 (65%) of 107 patients in the standard-dose UFH group, 63 (58%) of 108 in the low-dose UFH group (absolute risk difference -7.1 percentage points [95% CI -20.0 to 5.9]), and 62 (59%) of 105 in the LMWH group (-6.4 percentage points [-19.4 to 6.7]). Among 195 patients with a severe bleeding complication, 100 (51%) subsequently died. Severe thromboembolic complications occurred in 12 (11%) patients in the standard-dose UFH group, 11 (10%) in the low-dose UFH group, and nine (9%) in the LMWH group. The absolute risk difference compared with standard-dose UFH was -1.0 percentage points (95% CI -9.3 to 7.2) with low-dose UFH and -2.6 percentage points (-10.7 to 5.4) with LMWH. Among 32 patients with a severe thromboembolic complication, 17 (53%) died. At 6 months, 54 (50%) patients in the standard-dose UFH group, 45 (42%) in the low-dose UFH group, and 46 (44%) in the LMWH group had died. The absolute risk differences versus standard-dose UFH were -8.8 percentage points (95% CI -21.0 to 5.0) with low-dose UFH and -6.7 percentage points (-20.1 to 6.8) with LMWH.

Median ventilator-free days at day 60 were 0 (IQR 0–48) in the standard-dose UFH group, 15 (0–50) in the low-dose UFH group, and 9 (0–48) in the LMWH group (table 2). Median ECMO-free days were 0 (IQR 0–53) in the standard-dose UFH group, 39 (0–55) in the low-dose UFH group, and 44 (0–54) in the LMWH group. Median ICU-free days were 0 (IQR 0–41) in the standard-dose UFH group, 8 (0–42) in the low-dose UFH group, and 0 (0–40) in the LMWH group. The median value of 0 days primarily reflects the imbalanced ranking score due to the fact that more than 50% of the patients died before day 60 and hence had 0 days alive and free from support at day 60.

Median red blood cell, plasma, and platelet transfusion rates did not differ between groups (appendix p 2). The mean number of oxygenator exchanges per ECMO day was lower in the LMWH group than in the standard-dose UFH group (0.01 [SD 0.02] vs 0.02 [0.11]; $p=0.037$), with no significant difference between low-dose and standard-dose UFH.

	Standard-dose UFH group (n=107)	Low-dose UFH group (n=108)	LMWH group (n=105)
Sex			
Male	70 (65%)	79 (73%)	76 (72%)
Female	37 (35%)	29 (27%)	29 (28%)
Age, years	55 (43–64)	56 (45–64)	58 (47–68)
Ethnicity			
White	85 (79%)	89 (82%)	81 (77%)
Asian	4 (4%)	3 (3%)	1 (1%)
Black	4 (4%)	5 (5%)	3 (3%)
Other	4 (4%)	3 (3%)	7 (7%)
Unknown	10 (9%)	8 (7%)	13 (12%)
BMI, kg/m ²	27.8 (23.0–31.4)	27.9 (24.7–31.5)	28.1 (24.7–32.7)
APACHE IV score*	61 (48–89)	70 (52–90)	73 (55–96)
SOFA score†	11 (8–13)	11 (9–14)	12 (8–13)
ECMO blood flow, L/min	3.38 (0.68)	3.45 (0.75)	3.38 (0.73)
ECMO mode			
Veno-venous	46 (43%)	40 (37%)	42 (40%)
Veno-arterial	61 (57%)	68 (63%)	63 (60%)
ECMO indication			
Pulmonary	47 (44%)	43 (40%)	45 (43%)
Cardiac	50 (47%)	55 (51%)	49 (47%)
Extracorporeal cardiopulmonary resuscitation	10 (9%)	10 (9%)	11 (11%)
COVID-19	15 (14%)	17 (16%)	14 (13%)
Cannulation method			
Central	15 (14%)	14 (13%)	12 (11%)
Peripheral	92 (86%)	94 (87%)	93 (89%)
Cannula			
Heparin coated	85 (79%)	90 (83%)	86 (82%)
Non-heparin coated	22 (21%)	18 (17%)	19 (18%)
ECMO equipment			
Cardiohelp	89 (83%)	91 (84%)	89 (84%)
Xenios	11 (10%)	13 (12%)	11 (11%)
Other	7 (7%)	4 (4%)	5 (5%)
Antiplatelets			
Aspirin	8 (8%)	14 (13%)	6 (6%)
Cangrelor	0	3 (3%)	4 (4%)
Clopidogrel	5 (5%)	4 (4%)	10 (10%)
Ticagrelor	5 (5%)	6 (6%)	7 (7%)

Data are n (%), median (IQR), or mean (SD). APACHE=Acute Physiology and Chronic Health Evaluation. ECMO=extracorporeal membrane oxygenation. LMWH=low-weight-molecular heparin. SOFA=Sequential Organ Failure Assessment. UFH=unfractionated heparin. *APACHE IV score on the day of admission to the intensive care unit, with higher values indicating higher mortality risk. †SOFA score on the day of ECMO cannulation (higher scores indicate more severe organ dysfunction and a correspondingly higher risk of mortality in the intensive care unit).

Table 1: Baseline characteristics

Predefined subgroup analyses of the primary outcome and its components are shown in the appendix (pp 7–10). Point estimates showed relative risks for 6-month mortality in the low-dose UFH group among patients receiving veno-venous ECMO of 0.49 (95% CI 0.26–0.95) and among patients younger than 56 years of 0.54 (0.31–0.94), compared with the standard-dose UFH group (appendix p 8).

Crossovers to a different anticoagulation regimen occurred in 23 (22%) of 107 patients in the standard-dose UFH group, 14 (13%) of 108 in the low-dose UFH group, and 29 (28%) of 105 in the LMWH group (appendix p 3). Among patients who crossed over, few did so before the composite endpoint was reached: one in the standard-dose UFH group, five in the low-dose UFH group, and 13 in the LMWH group. The composite endpoint subsequently occurred in most of these patients (appendix p 3). The per-protocol population included 310 patients and yielded results consistent with the primary intention-to-treat analysis (appendix p 3).

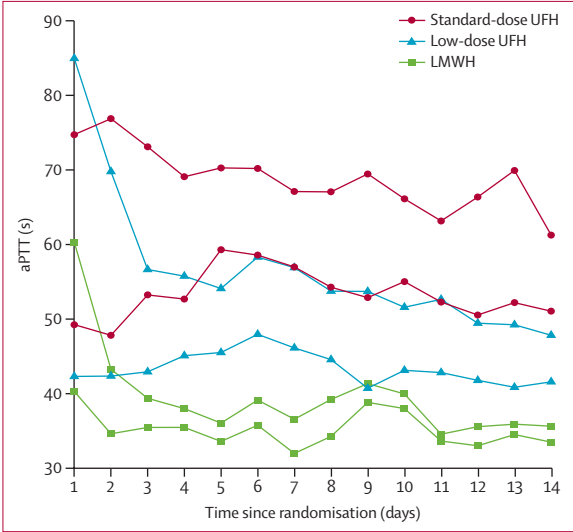


Figure 2: Lowest and highest aPTT values per day
aPPT=activated partial thromboplastin time. LMWH=low-molecular-weight heparin. UFH=unfractionated heparin.

At 6 months, median EQ-5D-5L utility scores in survivors were 0·80 (IQR 0·59–0·89) in the standard-dose UFH group, 0·83 (0·65–0·89) in the low-dose UFH group, and 0·79 (0·66–0·89) in the LMWH group; corresponding median visual analogue scale scores were 70 (IQR 60–80) in the standard-dose group, 72 (60–80) in the low-dose UFH group, and 70 (60–85) in the LMWH group (table 2). Incremental utilities were 0·07 for low-dose UFH and 0·01 for LMWH compared with standard-dose UFH.

Mean total costs were €127 601 (95% CI 107 565–150 463) in the standard-dose UFH group, €118 890 (102 685–136 904) in the low-dose UFH group, and €119 866 (103 561–137 679) in the LMWH group. The ICER was –€124 111 (95% CI –2145 607 to 1894 645) for low-dose UFH versus standard-dose UFH and –€523 131 (–3 346 890 to 3 954 755) for LMWH versus standard-dose UFH.

In the cost-effectiveness planes, 64% of bootstrap replicates for low-dose UFH and 41% for LMWH were in the southeastern quadrant, indicating better outcomes and lower costs (appendix p 13). Both intervention groups had a higher probability of being cost-effective compared with control across a range of willingness-to-pay thresholds (€0 to €50 000), with the highest probability observed for low-dose UFH (73–81%), whereas LMWH showed 72–74% probability of being cost-effective. Cost-effectiveness acceptability curves can be found in the appendix (pp 15–16).

Discussion

In this multicentre, randomised trial involving adults receiving ECMO, both low-dose UFH and therapeutic LMWH were non-inferior to standard-dose UFH for the composite outcome of severe bleeding, severe thromboembolic complications, and 6-month mortality.

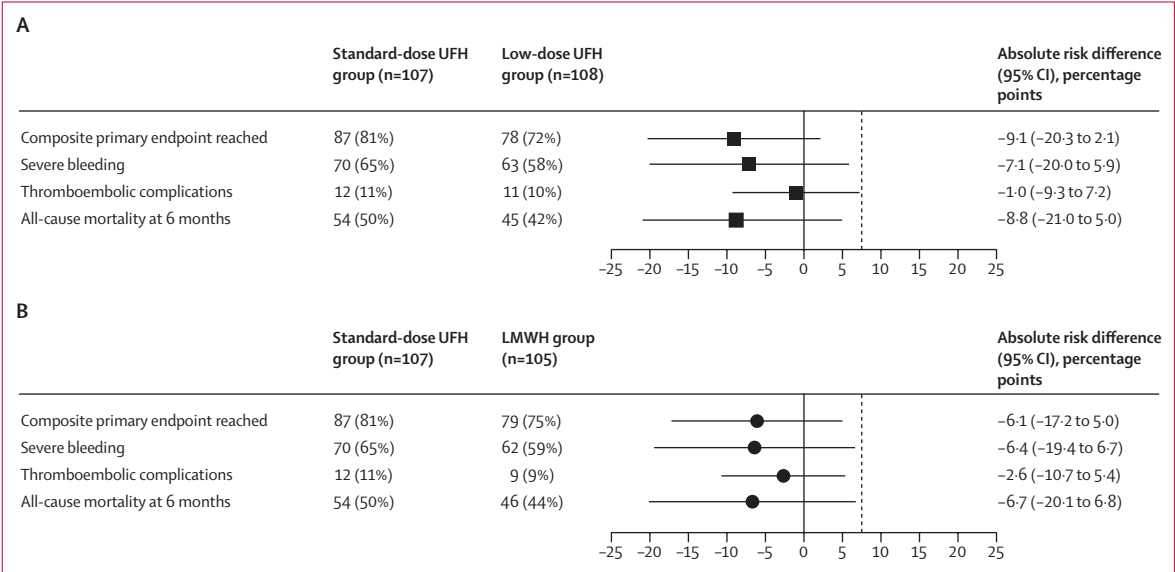


Figure 3: Absolute risk difference of primary composite endpoint
LMWH=low-molecular-weight heparin. UFH=unfractionated heparin.

	Standard-dose UFH group (n=107)	Low-dose UFH group (n=108)	p value	LMWH group (n=105)	p value
Reason for ECMO discontinuation	..	..	0.41	..	0.023
Expected recovery	60 (56%)	72 (66%)	..	73 (70%)	..
Died on extracorporeal life support	29 (27%)	22 (20%)	..	20 (19%)	..
Diagnosis incompatible with life	6 (6%)	6 (6%)	..	10 (10%)	..
Left ventricular assist device, heart transplant, or lung transplant	9 (8%)	4 (4%)	..	2 (2%)	..
ECMO complication	3 (3%)	4 (4%)	..	0	..
Tracheostoma	8 (8%)	16 (15%)	0.083	21 (20%)	0.0070
Alive at discharge	58 (54%)	66 (61%)	0.31	60 (57%)	0.67
Discharge location	..	..	0.90	..	0.41
Home	20/58 (34%)	20/62 (32%)	..	23/60 (38%)	..
Referral hospital	12/58 (21%)	15/62 (24%)	..	7/60 (12%)	..
Other	26/58 (45%)	27/62 (44%)	..	30/60 (50%)	..
Mechanical ventilation duration, days	13 (6–29)	10 (5–20)	0.27	13 (7–23)	0.98
ICU length of stay, days	16 (7–33)	13 (8–25)	0.38	18 (8–34)	0.45
Hospital length of stay, days	41 (19–64)	35 (20–52)	0.35	36 (23–60)	0.56
ECMO duration, days	7 (3–12)	8 (4–13)	0.61	6 (4–12)	0.69
EQ-5D-5L, index	0.80 (0.59–0.89)	0.83 (0.65–0.89)	0.50	0.79 (0.66–0.89)	0.70
EQ-5D-5L, VAS	70 (60–80)	72 (60–80)	0.44	70 (60–85)	0.36
Alive at day 60, days					
Without ECMO	0 (0–53)	39 (0–55)	0.45	44 (0–54)	0.24
Without mechanical ventilation	0 (0–48)	15 (0–50)	0.53	9 (0–48)	0.69
ICU free	0 (0–41)	8 (0–42)	0.25	0 (0–40)	0.61

Data are n (%) or median (IQR). Percentages might not sum to 100 because of rounding. ECMO=extracorporeal membrane oxygenation. ICU=intensive care unit. LMWH=low-molecular-weight heparin. UFH=unfractionated heparin. VAS=visual analogue scale.

Table 2: Secondary outcomes

Event rates were lower in both intervention groups, driven primarily by a reduction in bleeding complications, without an increase in thromboembolic events or mortality. Given the association between reduced bleeding, lower transfusion needs, and lower resource use, these findings are consistent with potential cost-effectiveness while maintaining clinical safety.

Our findings build on previous observational studies and a small phase 2 randomised trial that suggested lower-intensity anticoagulation might reduce bleeding without increasing thrombotic complications.^{24,33,34} Large registry analyses have consistently shown that bleeding complications are more frequent and more strongly associated with mortality than thrombotic events, underscoring the rationale for reduced-intensity strategies.¹⁹ Modern ECMO systems with centrifugal pumps and biocompatible circuits have further decreased the incidence of clinically significant thrombosis, diminishing the need for intensive anticoagulation.⁸ In our trial, the overall incidence of the primary outcome and its individual components was higher than anticipated at the design stage, which might reflect more complete ascertainment of events and the use of broad, standardised definitions. Still, incidences were similar across all three study groups and similar to rates reported in previous randomised and observational studies.^{13,15,34} Taken together,

these findings highlight that bleeding remains common in contemporary ECMO, probably driven in part by high-intensity anticoagulation, and underscore the persistent burden of bleeding, thrombotic events, and mortality despite advances in ECMO technology.^{35–41} This reinforces the need for evidence-based strategies to optimise anticoagulation and minimise harm, to which the present trial provides the first adequately powered contribution.

Although both intervention strategies were non-inferior to standard-dose UFH, some differences merit consideration. Low-dose UFH was associated with fewer bleeding events and with lower mortality than standard-dose UFH in some subgroups. These differences might be related to reduced transfusion exposure, a factor previously linked to adverse outcomes during ECMO.^{14,16,42,43} However, these subgroup findings should be interpreted cautiously, given the non-inferiority design, multiple testing, and absence of formal type I error control. No formal interaction testing was performed; therefore, differences between subgroups should not be interpreted as evidence of heterogeneity of treatment effect. No major heterogeneity across subgroups was apparent, although small sample sizes limited these analyses and they should be considered exploratory. LMWH showed the lowest transfusion requirements and fewer oxygenator exchanges, findings that are consistent with its more predictable

pharmacokinetic profile. Notably, both regimens demonstrated similar safety relative to standard-dose UFH, supporting low-dose anticoagulation as a viable alternative in both veno-venous and veno-arterial ECMO.

Strengths of this trial include its randomised, multicentre design, adequate power, prespecified endpoints, complete 6-month follow-up, and inclusion of both veno-venous and veno-arterial ECMO. A further strength of the trial is faster-than-anticipated enrolment, leading to earlier than planned completion of the study, and lower-than-expected loss to follow-up, leading to more patients eligible for analyses. Our study is also the first randomised trial to evaluate therapeutic LMWH during ECMO; previous data were limited to small observational series with prophylactic dosing and a small feasibility study in 26 patients.⁴⁴ LMWH offers advantages such as more predictable anticoagulation and fewer monitoring requirements, making it a reasonable alternative to UFH.⁴⁵

Although combining veno-venous and veno-arterial ECMO in one study introduces substantial pathophysiological heterogeneity, the underlying rationale of the trial applies to both modalities. Contemporary anticoagulation targets are probably higher than necessary, and lowering anticoagulation intensity might mitigate bleeding and thereby improve outcomes. Prespecified subgroup analyses yielded consistent findings across veno-venous and veno-arterial ECMO, supporting the robustness and generalisability of the results. Although a few centres already use lower anticoagulation targets in veno-venous ECMO, our findings demonstrate the feasibility of reduced-intensity anticoagulation in veno-arterial ECMO, in which evidence has been limited and physicians have been more reluctant to adopt low-dose strategies.³³

Contemporary practice patterns seem to have already shifted towards lower anticoagulation targets compared with historical guideline recommendations. The international prospective PROTECMO study across 8471 ECMO days in 41 centres reported a median aPTT of 52 s in patients receiving veno-venous ECMO with a broad IQR of 39–61, and these values include aPTT values that decreased on average by 5·3 s after a bleeding event, which occurred in 53% of the patients.² Similar observational analyses have suggested that targeting absolute aPTT values around 45–55 s might reduce bleeding without an increase in thrombosis. In the absence of randomised trials, there is still clinical equipoise regarding the optimal anticoagulation target. Our trial now provides prospective randomised data to support the safety and clinical appropriateness of this ongoing international practice change.

Although we used aPTT as guidance for anticoagulation strategy in our study, its interpretation in patients supported with ECMO has recognised limitations. aPTT values can be influenced by factors unrelated to anticoagulant effects, including factor deficiencies, inflammatory states, and the presence of lupus anticoagulant, which might reduce reliability as a surrogate

for heparin activity. These limitations are increasingly recognised in ECMO populations and have contributed to growing interest in alternative monitoring strategies such as anti-factor Xa-guided anticoagulation. Thereby, aPTT values were presented in absolute seconds, which might limit comparability across centres because reagent-specific differences can affect sensitivity to UFH. However, analyses focused on within-trial comparisons between randomised groups rather than absolute thresholds, mitigating the impact of inter-laboratory variability. Furthermore, aPTT monitoring is not relevant for LMWH, and observed values in the LMWH group were only mentioned to show deviations between allocation groups. Nevertheless, aPTT remains widely used in clinical practice and was chosen to reflect contemporary standard care.⁴⁶ However, when implementing the results of our study, every centre must be aware of the variability in the sensitivity of aPTT reagents to UFH.

The effectiveness of anticoagulation strategies might depend not only on target ranges but also on the quality of anticoagulation management, including time to reach target and time spent within range. Our study had a pragmatic design in which UFH dosing and adjustment followed local institutional protocols. Although this approach enhances external validity and reflects real-world practice, it might have introduced variability in anticoagulation management across centres and subsequent variability in time to target and time within the target range. These parameters were not analysed in detail. However, figure 2 provides an overview of the aPTT trajectory over time across the three groups, offering an illustrative insight into the overall course of anticoagulation. In addition, we found a clear separation of aPTT values between both groups (from the first timepoint), which further increases confidence in adequate group separation.

A proportion of patients crossed over to another anticoagulation regimen, but only a subset did so before the composite endpoint was reached. Among these patients, the composite endpoint was subsequently reached in most cases, indicating that crossovers predominantly occurred in the context of clinical deterioration. As such, these crossovers were unlikely to have affected the primary outcome and mainly reflected protocol-driven management decisions. Results of the per-protocol analysis were consistent with the primary analysis. Our trial used aPTT exclusively to monitor UFH, reflecting its wide availability, low cost, and feasibility in a multicentre setting.^{15,47,48} Anti-factor Xa activity was recorded midway through the trial for study purposes but was not used for dose adjustment. Although recent guidelines recommend anti-factor Xa as the preferred monitoring strategy because of its closer correlation with heparin dose, the observed safety of low-dose UFH in our trial is likely to be relevant also for centres using anti-factor Xa-based protocols. An anti-factor Xa target of 0·2–0·3 IU/mL might therefore be appropriate, although this extrapolation should be interpreted with caution.

Because routine screening for venous thromboembolic events during or after ECMO support is not customary in the participating centres, the incidence of such events in our cohort might have been underestimated.⁴⁹ This limitation warrants consideration when interpreting these findings. However, the clinical relevance of such unrecognised events remains uncertain and might be limited.

The multiplicity inherent to multiple comparisons represents an important limitation of the study design. Two primary treatment comparisons, an interim analysis, and numerous secondary endpoints increase the potential for type I error inflation. Consistent with contemporary trial methodology, formal control of type I error was focused on the primary endpoint. No adjustments were applied to secondary endpoints, and these analyses should therefore be interpreted as exploratory and hypothesis generating. The non-inferiority margin was prespecified on the basis of clinical judgement, anticipated event rates, and feasibility considerations in a population of critically ill patients. Interpretation of non-inferiority findings should consider effect estimates and CIs rather than the margin alone. The open-label design and variability in local monitoring routines for aPTT measurement and UFH dose adjustment are further limitations. However, the primary endpoint was based on objective clinical events, which limits the potential for bias. Detailed screening information distinguishing between missed patients and those deemed to be ineligible was not systematically collected, limiting assessment of potential selection bias during patient inclusion. Another limitation is that baseline EQ-5D-5L data were not collected, and QALYs were therefore estimated without adjustment for baseline utility. Although randomisation is expected to balance baseline characteristics, this assumption cannot be verified, and guidelines recommend adjustment for baseline utility. This might have introduced some uncertainty in the QALY estimates. Another limitation of this study is the absence of data on achieved anticoagulation intensity, such as time to reach and time spent within the therapeutic range. Although these parameters provide important mechanistic insight into anticoagulation quality, their assessment would introduce post-randomisation variables that might be affected by treatment allocation and clinical course, potentially introducing bias. Future analyses are warranted to further explore these aspects.

In conclusion, among patients receiving ECMO, both low-dose UFH and therapeutic LMWH were non-inferior to standard-dose UFH for a composite of major bleeding, major thrombotic events, and 6-month mortality. The economic evaluation indicated lower costs with preserved or slightly improved quality of life. These findings support reconsideration of current anticoagulation targets in ECMO practice.

Dutch ECLS Study Group

Maite M T van Haeren (Department of Intensive Care Medicine, Amsterdam University Medical Center, Amsterdam, Netherlands),

Matty Koopmans (Department of Intensive Care, OLVG, Amsterdam, Netherlands), C L Meuwese (Department of Intensive Care Medicine, Erasmus Medical Center, Rotterdam, Netherlands), Marcella C A Müller (Department of Intensive Care Medicine, Amsterdam University Medical Center, University of Amsterdam, Amsterdam, Netherlands), Dirk W Donker (Department of Critical Care, University Medical Center Utrecht, Utrecht University, Utrecht, Netherlands; Cardiovascular and Respiratory Physiology Group, TechMed Centre, University of Twente, Enschede, Netherlands), Sakir Akin (Department of Intensive Care, Haga Hospital, The Hague, Netherlands), Judith van den Brule (Department of Intensive Care Medicine, Radboud University Medical Centre, Nijmegen, Netherlands).

Contributors

OvM, AOL-H, and WMvdb conceived and designed the study, had full access to all the data, and verified the underlying data. OvM drafted the manuscript. KMV conducted the cost-effectiveness analysis and contributed to drafting the corresponding sections of the manuscript. PV verified the statistical analyses. All other authors contributed to data acquisition and interpretation, critically revised the manuscript for important intellectual content, and approved the final version. OvM, AOL-H, and WMvdb had final responsibility for the decision to submit for publication.

Declaration of interests

We declare no competing interests.

Data sharing

The study data are available from the corresponding author upon reasonable request after an embargo of 6 months after publication.

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